AI CIRCUIT EXPLAINER AND VERIFICATION STUDIO

ECE Capstone Project Report

Project question
Can AI correctly explain and analyse basic analog circuits, and how can its answers be checked with simulation?

Prepared by: _____
Register number: _____
Department / College: _____
Academic year: _____

Prepared as a complete working report template. Replace the student details and attach your own simulator screenshots and exported measurements before final submission.

Abstract and project overview

This project studies eight introductory analog circuits and uses AI assistants to explain their operation, derive key equations, and estimate expected outputs. The calculations are then checked independently using circuit theory and a circuit simulator such as LTspice or Falstad. The project also records common AI failure modes, compares four AI assistants using a repeatable evaluation rubric, and demonstrates a simple automated comparison workflow.

The central engineering principle is that an AI-generated answer is a hypothesis, not a measurement. Component values, circuit topology, model assumptions, units, operating regions, and simulator settings must be checked by the student or engineer.

Objectives

1. Explain eight basic analog circuits at beginner and advanced levels.
2. Calculate important values such as gain, cutoff frequency, output voltage, and bias current.
3. Establish independent hand-calculation reference values.
4. Prepare simulator test plans and record measured results.
5. Compare AI assistants using identical prompts and a documented rubric.
6. Build a repeatable calculation-versus-simulation comparison and error log.
7. Explain AI limitations and the need for engineering verification.

Scope and assumptions

The numerical examples use ideal or first-order models unless stated otherwise. Real components have tolerances, temperature dependence, parasitic effects, and non-ideal characteristics. Simulator results will vary with device models, op-amp selection, diode models, load values, and solver settings. Values marked “expected” are analytical predictions, not fabricated simulator measurements.

1. Methodology and verification workflow

Use the same workflow for every circuit so the AI comparison is fair and repeatable.

Stage	Action	Evidence to save
1. Define	Draw the topology; list supply, component values, and load.	Circuit sketch and input data
2. Ask AI	Use the same prompt with at least four AI tools.	Prompt and unedited responses
3. Check theory	Derive the result independently from circuit laws.	Hand calculation
4. Simulate	Build the same circuit in LTspice or Falstad; run the correct analysis.	Schematic and plot screenshots
5. Compare	Record AI, hand-calculated, and simulator values.	Comparison table
6. Diagnose	Investigate differences, units, assumptions, and model effects.	Error log and corrected answer
7. Report	Explain what matched, what did not, and why.	Verified explanation

Suggested simulator setup

LTspice: draw the circuit, assign component values, add ground, set the input source, then use .op for DC operating points, .ac for frequency response, or .tran for time-domain waveforms. Label important nodes and save the schematic and plot window.

Falstad: build the circuit in the browser simulator, set component values and source parameters, enable scope traces, and capture the circuit plus waveform. Ensure the displayed units and time/frequency scale are readable.

Important: No simulator has been run as part of preparing this PDF. The “expected” results below are theory-based targets. Insert your own screenshots and actual simulator readings in the marked evidence section, then update the comparison table.

2.1 Resistive voltage divider

Beginner explanation: Two series resistors divide an input voltage. The output is taken across the lower resistor. It is used for reference voltages and signal scaling, but the output changes if the load draws significant current.

Given values: Vin = 12 V; R1 = 10 kΩ; R2 = 10 kΩ; output across R2; no load.

Hand calculation: Vout = Vin × R2/(R1+R2) = 12 × 10/(10+10) = 6.00 V.
I = Vin/(R1+R2) = 12/20 kΩ = 0.600 mA.
Power in each resistor = I^2R = 3.60 mW.

Expected result: 6.00 V DC at the output; 0.600 mA branch current.

Simulator verification: Use .op analysis. Probe Vin and Vout. Add a load resistor only as a separate test; a load in parallel with R2 changes the divider ratio.

Common AI / engineering error: A frequent AI mistake is to use R1 in the numerator even though the output is measured across R2. Always identify the output terminals before selecting the formula.

Evidence to attach: schematic screenshot; output plot or operating-point screenshot; measurement cursor or operating-point values; simulator name and version.

2.2 RC low-pass filter

Beginner explanation: A series resistor and shunt capacitor pass low frequencies and attenuate high frequencies. The capacitor impedance decreases as frequency rises, pulling the output toward ground.

Given values: R = 1.0 kΩ; C = 0.10 µF; Vin = 5 V amplitude; test frequency = 1 kHz.

Hand calculation: fc = 1/(2πRC) = 1/[2π(1000)(0.10×10■■)] = 1,591.5 Hz.
|H| = 1/√[1+(f/fc)²] = 0.8467 at 1 kHz.
Vout = 5 × 0.8467 = 4.23 V amplitude. Phase = –tan■¹(f/fc) ≈ –32.1°.

Expected result: At 1 kHz, magnitude ≈ 0.847 and output ≈ 4.23 V amplitude. At fc, magnitude is 0.707 (–3 dB).

Simulator verification: Use .ac analysis from about 10 Hz to 100 kHz. Plot V(out)/V(in), locate the –3 dB point, and compare it with 1.59 kHz.

Common AI / engineering error: Do not confuse 0.1 µF with 0.1 F. A unit error changes the cutoff frequency by a factor of one million.

Evidence to attach: schematic screenshot; output plot or operating-point screenshot; measurement cursor or operating-point values; simulator name and version.

2.3 RC high-pass filter

Beginner explanation: A series capacitor and shunt resistor pass high frequencies and attenuate low frequencies. At very low frequency the capacitor blocks the signal; at high frequency it behaves approximately as a short.

Given values: R = 1.0 kΩ; C = 0.10 µF; Vin = 5 V amplitude; test frequency = 1 kHz.

Hand calculation: fc = 1/(2πRC) = 1,591.5 Hz.
|H| = (f/fc)/√[1+(f/fc)²] = 0.5320 at 1 kHz.
Vout = 5 × 0.5320 = 2.66 V amplitude. Phase = 90° – tan■¹(f/fc) ≈ +57.9°.

Expected result: At 1 kHz, magnitude ≈ 0.532 and output ≈ 2.66 V amplitude. At fc, magnitude is 0.707.

Simulator verification: Use .ac analysis. Verify that gain rises with frequency and approaches unity well above cutoff.

Common AI / engineering error: The high-pass transfer function is not the same as the low-pass expression. Check the low-frequency and high-frequency limits to catch a swapped formula.

Evidence to attach: schematic screenshot; output plot or operating-point screenshot; measurement cursor or operating-point values; simulator name and version.

2.4 BJT common-emitter amplifier (voltage-divider bias)

Beginner explanation: A common-emitter transistor stage can amplify a small input signal and invert its phase. The base divider sets the DC bias, while RC and RE influence collector current and voltage gain.

Given values: VCC = 12 V; R1 = 47 kΩ (VCC to base); R2 = 10 kΩ (base to ground); RC = 2.2 kΩ; RE = 1.0 kΩ; β = 100; VBE = 0.70 V; assume forward-active region.

Hand calculation: VTH = 12×10/(47+10) = 2.105 V; RTH = 47k||10k = 8.246 kΩ.
IB ≈ (VTH–VBE)/[RTH+(β+1)RE] = 12.85 µA.
IC ≈ βIB = 1.285 mA; IE ≈ 1.298 mA.
VE ≈ 1.298 V; VC = 12–IC×2.2k = 9.17 V; VCE ≈ 7.88 V.

Expected result: Approximate DC Q-point: IC ≈ 1.29 mA and VCE ≈ 7.9 V. This indicates a plausible forward-active operating point under the stated model.

Simulator verification: Use .op with a suitable NPN model (for example, a small-signal general-purpose NPN). Record IB, IC, VB, VE, VC, and VCE. For gain, add a small AC input and run .ac or transient analysis.

Common AI / engineering error: β varies between devices and with current. The simple constant-β model is approximate. Do not claim an exact gain from the DC bias calculation; gain also depends on load, frequency, bypass capacitors, and transistor model.

Evidence to attach: schematic screenshot; output plot or operating-point screenshot; measurement cursor or operating-point values; simulator name and version.

2.5 Op-amp inverting amplifier

Beginner explanation: The inverting amplifier uses negative feedback. The non-inverting input is grounded, and the feedback network keeps the inverting input near virtual ground while the amplifier remains in its linear region.

Given values: Ideal op-amp; Rin = 10 kΩ; Rf = 100 kΩ; Vin = +0.20 V DC; supply rails sufficiently large (for example ±12 V).

Hand calculation: Av = –Rf/Rin = –100k/10k = –10.
Vout = Av×Vin = –10×0.20 = –2.00 V.

Expected result: Gain = –10 V/V; output = –2.00 V for +0.20 V input, assuming no saturation.

Simulator verification: Use an ideal op-amp or a real model with suitable supply rails. Run .op for DC and .tran for a small sine input. Confirm output inversion and gain magnitude.

Common AI / engineering error: The minus sign is essential. Also check output swing: a real op-amp cannot necessarily reach its supply rails.

Evidence to attach: schematic screenshot; output plot or operating-point screenshot; measurement cursor or operating-point values; simulator name and version.

2.6 Full-wave bridge rectifier

Beginner explanation: Four diodes form a bridge so both halves of an AC input produce the same output polarity. Two diodes conduct in each half-cycle. A capacitor filter can reduce ripple but is not included in the basic calculation.

Given values: Input = 12 V RMS sine at 50 Hz; silicon diode drop ≈ 0.70 V each; no smoothing capacitor; light load approximation.

Hand calculation: $V_{peak} = 12\sqrt{2} = 16.97$ V.
Approximate output peak ≈ 16.97–2(0.70) = 15.57 V.
Approximate average full-wave output ≈ $2(V_{peak}-1.4)/\pi = 9.91$ V.
Ripple frequency = 2×50 = 100 Hz.

Expected result: Expected waveform is full-wave rectified; peak near 15.6 V under the assumed constant-drop approximation; pulse repetition is 100 Hz.

Simulator verification: Run transient analysis for several AC cycles. Probe the source and load. Verify both half-cycles appear positive at the load. Add a capacitor only in a separate filtered test.

Common AI / engineering error: The two-diode drop applies to the conducting path. Do not subtract four diode drops. The average estimate is approximate and depends on load and diode model.

Evidence to attach: schematic screenshot; output plot or operating-point screenshot; measurement cursor or operating-point values; simulator name and version.

2.7 Symmetrical diode clipper

Beginner explanation: A pair of oppositely oriented diodes can limit a signal in both polarities. With silicon diodes, the output is approximately limited near +0.7 V and –0.7 V once the diodes conduct, depending on the exact topology and source resistor.

Given values: Sine input = 5 V peak; two antiparallel silicon diodes across output; series limiting resistor = 1 kΩ; approximate diode threshold = 0.70 V.

Hand calculation: For |Vin| below about 0.7 V, diodes are mostly off and Vout follows Vin. For larger positive or negative excursions, one diode conducts and clamps the output near ±0.7 V. Exact clamp levels depend on current and diode model.

Expected result: Expected output is a sine wave flattened near +0.7 V and –0.7 V; it is not a perfect hard clamp.

Simulator verification: Use transient analysis with a 5 V peak sine source. Plot Vin and Vout. Check diode current and output flattening around both polarities.

Common AI / engineering error: A diode clipper must include a current-limiting path. The diode orientation determines whether clipping is positive, negative, or symmetrical.

Evidence to attach: schematic screenshot; output plot or operating-point screenshot; measurement cursor or operating-point values; simulator name and version.

2.8 Zener shunt regulator

Beginner explanation: A reverse-biased Zener diode connected across a load can hold the load voltage near its Zener voltage while the series resistor limits current. Regulation is maintained only while the Zener current stays within its operating range.

Given values: Vin = 12 V DC; VZ = 5.1 V; RS = 330 Ω; RL = 1.0 kΩ; idealized Zener approximation.

Hand calculation: Load current IL = 5.1/1k = 5.10 mA.
Series current IS = (12–5.1)/330 = 20.91 mA.
Zener current IZ = IS–IL = 15.81 mA.
Zener power PZ = 5.1×15.81mA = 80.6 mW.
Resistor power PR = (6.9²)/330 = 144 mW.

Expected result: Expected Vout ≈ 5.1 V; Zener current ≈ 15.8 mA. Confirm the chosen Zener's power rating and minimum current.

Simulator verification: Use .op with a Zener model and record Vout, resistor current, and Zener current. Repeat with a heavier load to observe loss of regulation.

Common AI / engineering error: Never assume the output remains 5.1 V for every load. If the Zener current falls below its knee current, regulation degrades.

Evidence to attach: schematic screenshot; output plot or operating-point screenshot; measurement cursor or operating-point values; simulator name and version.

3. Calculation-versus-simulation verification table

The following values are analytical targets. Enter actual simulator measurements in the last column after running the circuits. The error column should be calculated from the measured result, not guessed.

Circuit / quantity	Hand-calculated target	Simulator result (fill in)	% error
Voltage divider Vout	6.00 V DC	_____	_____
RC low-pass |Vout| at 1 kHz	4.23 V amplitude	_____	_____
RC high-pass |Vout| at 1 kHz	2.66 V amplitude	_____	_____
BJT amplifier IC / VCE	1.29 mA / 7.88 V	_____	_____
Inverting op-amp Vout	–2.00 V	_____	_____
Bridge rectifier peak / average	15.57 V / 9.91 V approx.	_____	_____
Diode clipper clamp	about ±0.7 V	_____	_____
Zener regulator Vout / IZ	5.10 V / 15.81 mA approx.	_____	_____

Percentage error formula

% error = |(measured – reference)/reference| × 100%. For a reference value near zero, use absolute error or a suitable full-scale/normalised error instead. State the sign convention and units.

Interpretation guide

A small difference may result from rounding or non-ideal device models. A larger difference should trigger checks of topology, component units, source amplitude convention (peak versus RMS), load connection, diode model, transistor β, op-amp supply rails, and simulator analysis settings. Do not alter the hand calculation merely to force agreement; explain the physical reason for any justified difference.

4. Error log and correction record

Use this log to document actual mistakes found in AI responses or in your own work. The sample entries below are illustrative checks to look for; mark them as observed only if they occur in your testing.

ID	Potential error	How to detect it	Corrective action
E01	Divider output uses wrong resistor in numerator.	Check which resistor is connected from Vout to ground.	Use Vout = Vin×Rbottom/(Rtop+Rbottom).
E02	RC time constant uses µF as F.	Check dimensions of RC; seconds must result.	Convert µF to ×10■■ F before calculating.
E03	Low-pass formula used for high-pass circuit.	Check limiting behaviour at f→0 and f→∞.	Use the transfer function matching the topology.
E04	Bridge calculation subtracts four diode drops.	Trace the current path through the bridge.	Two diodes conduct in series in each half-cycle.
E05	Op-amp output sign omitted.	Check the inverting input and feedback topology.	Use Av = −Rf/Rin.
E06	BJT result assumes β is exact.	Compare ideal hand model with selected transistor model.	Report the assumed β and use simulator operating point.
E07	Zener assumed regulated under any load.	Check whether calculated IZ remains above knee current.	Recalculate current budget for each load.
E08	Peak and RMS values mixed.	Inspect source definition and waveform measurement.	Convert Vpeak = √2 Vrms for a sine wave.

Actual observation record (duplicate this table for each issue)

Date / circuit	AI tool / prompt	Original incorrect output	Verified correction and evidence
_____	_____	_____	_____
_____	_____	_____	_____

5. AI tool comparison

Compare at least four assistants with identical inputs. Suggested tools are ChatGPT, Google Gemini, Claude, and Perplexity. Product features and model versions change, so record the exact model/version, date, and settings used. The table below is a comparison plan, not a claim that these systems were tested for this report.

Tool	What to evaluate	Record during your test
ChatGPT	Clarity of step-by-step explanation; equation selection; ability to revise after correction.	Model/version, prompt, response, factual errors, clarity score.
Google Gemini	Circuit reasoning; handling of diagrams; unit consistency; explanation quality.	Model/version, prompt, response, factual errors, clarity score.
Claude	Careful assumptions; mathematical consistency; error acknowledgement; structured report quality.	Model/version, prompt, response, factual errors, clarity score.
Perplexity	Source transparency where references are used; explanation quality; equation correctness.	Model/version, prompt, sources, factual errors, clarity score.

Repeatable scoring rubric

Score each category from 1 (poor) to 5 (excellent): circuit interpretation, correct equations, numerical accuracy, unit handling, explanation clarity, acknowledgement of assumptions, and correction after feedback. Total maximum = 35. Do not rank tools based on one prompt alone; test at least three circuits and retain the raw responses.

AI tool	Interpretation /5	Equations /5	Numerics /5	Units /5	Clarity /5	Assumptions /5	Correction /5	Total /35
ChatGPT	__	__	__	__	__	__	__	__
Gemini	__	__	__	__	__	__	__	__
Claude	__	__	__	__	__	__	__	__
Perplexity	__	__	__	__	__	__	__	__

Fair-test rules

Use identical circuit values and wording; start a fresh chat for each tool; do not show one tool another tool's answer; ask for assumptions and units; preserve the first response before requesting corrections; and use the same independent reference calculation for scoring.

6. Prompt library

Copy and reuse these prompts. Replace the bracketed fields with the exact circuit details. Save the prompt and full response for each AI tool.

P1 — Beginner explanation

Explain this analog circuit to a first-year ECE student. Describe the function of every component, current paths, input/output behaviour, and one practical application. Circuit: [topology]. Values: [complete values]. State all assumptions and do not invent missing values.

P2 — Advanced analysis

Analyse the following circuit at an engineering level. Identify the correct operating mode, derive the governing equations, show substitutions with SI units, and calculate [requested quantities]. State approximations and conditions under which they are valid. Circuit: [topology and values].

P3 — Independent verification

Check the following calculation independently. Re-derive it from first principles rather than copying the supplied steps. Identify any incorrect formula, arithmetic, sign, unit, or assumption. Calculation: [paste calculation]. Circuit: [paste topology and values].

P4 — Simulator setup

Give a simulator setup plan for [LTspice/Falstad] for this circuit. Specify source waveform and amplitude convention, component values, ground, analysis type, sweep/time settings, nodes to probe, and expected qualitative waveform. Do not claim that you ran the simulator.

P5 — Error audit

Audit this answer for at least five possible failure modes: wrong topology, wrong formula, unit conversion, peak/RMS confusion, sign convention, device operating region, loading, and unrealistic ideal assumptions. Separate confirmed errors from items that require simulation.

P6 — Diagram validation

Review the attached schematic. Trace the current path and identify component polarity, pinout, grounding, and connection errors. If a connection is ambiguous, ask for clarification instead of guessing.

P7 — Beginner plus advanced response

Give two explanations of [circuit]: (A) beginner-level intuition in plain language; (B) advanced derivation with equations and assumptions. Use the same component values in both sections and ensure the numerical result agrees.

P8 — Comparison summary

Compare the AI result, hand calculation, and simulator measurement in this table: [paste table]. Calculate percentage error where appropriate, explain likely causes, and do not change measured values to make them match.

7. Automation: calculation-versus-simulation checker

A simple CSV-based workflow can compare reference calculations with simulator measurements. The script below reads a CSV containing circuit, reference value, measured value, and unit; it calculates absolute and percentage error and writes a new CSV report.

Example input file (measurements.csv)

```
circuit,reference,measured,unit
Voltage divider,6.00,5.98,V
Op-amp inverting,-2.00,-1.97,V
Zener regulator,5.10,5.06,V
```

Python automation code

```
import csv

input_file = "measurements.csv"
output_file = "comparison_report.csv"

with open(input_file, newline="") as f:
    rows = list(csv.DictReader(f))

results = []
for row in rows:
    reference = float(row["reference"])
    measured = float(row["measured"])
    absolute_error = abs(measured - reference)

    if abs(reference) > 1e-12:
        percent_error = absolute_error / abs(reference) * 100
        percent_text = f"{percent_error:.2f}"
    else:
        percent_text = "N/A (reference is zero)"

    results.append({
        "circuit": row["circuit"],
        "reference": reference,
        "measured": measured,
        "unit": row["unit"],
        "absolute_error": f"{absolute_error:.6g}",
        "percent_error": percent_text
    })

with open(output_file, "w", newline="") as f:
    fields = ["circuit", "reference", "measured", "unit",
              "absolute_error", "percent_error"]
    writer = csv.DictWriter(f, fieldnames=fields)
    writer.writeheader()
    writer.writerows(results)

print(f"Saved {output_file}")
```

How to use

1. Save the input data as measurements.csv in the same folder as the script.
2. Enter the actual simulator readings, not expected or invented readings.
3. Run the script using Python 3.
4. Open comparison_report.csv and check each unit and sign.
5. Explain large errors in the report. This automation checks arithmetic only; it cannot decide whether the circuit model or reference calculation is physically correct.

Example check (illustrative values only): if a divider reference is 6.00 V and simulator reading is 5.98 V, absolute error = 0.02 V and percentage error = 0.33%. Replace the sample measurements with your own results.

8. Verified study guide

Core concepts to remember

Ohm's law: V = IR. Use consistent SI units.
Series resistors: the same current flows through each component; resistances add.
Parallel resistors: voltage is common; reciprocal resistances add.
Capacitor impedance: ZC = 1/(jωC).
RC cutoff: fc = 1/(2πRC) for a first-order RC filter.
Decibels: voltage gain in dB = 20 log10(|Vout/Vin|).
BJT: in forward-active operation, IC is approximately βIB, but β is not a fixed precision value.
Ideal inverting op-amp: Av = –Rf/Rin, provided negative feedback is active and output is not saturated.
Rectification: a bridge rectifier uses two conducting diodes per half-cycle.
Zener regulator: series resistance limits current; the Zener must remain in its regulation region.

Common unit conversions

Quantity	Conversion	Example
kilo (k)	10^3	10 kΩ = 10,000 Ω
milli (m)	10■³	2 mA = 0.002 A
micro (μ)	10■■	0.1 μF = 1×10■■ F
nano (n)	10■■	100 nF = 1×10■■ F

Quick verification checklist

■ Is the schematic identical to the circuit being calculated?
■ Are all component values and units correct?
■ Is the output node measured at the correct location?
■ Is the source specified in peak, peak-to-peak, or RMS units?
■ Is the selected analysis appropriate (DC, AC, or transient)?
■ Are semiconductor models and supply rails realistic?
■ Are loading and initial conditions considered?
■ Are screenshots readable and labelled?
■ Are AI claims separated from independently verified facts?

9. Limitations of AI and engineering responsibility

AI assistants can produce fluent explanations while making subtle technical errors. They may select a formula for a different topology, silently assume ideal devices, confuse peak and RMS values, omit loading, misread a schematic, or perform arithmetic with incorrect units. They may also describe a simulator procedure as if a simulation had been run when no run occurred.

Why engineer verification is necessary

1. **Physical correctness:** circuit behaviour depends on real device characteristics and operating regions.
2. **Traceability:** engineering decisions need equations, assumptions, source values, and reproducible evidence.
3. **Safety and reliability:** incorrect component power or voltage estimates can damage hardware or create unsafe conditions.
4. **Context:** AI may not know the exact model, pinout, tolerance, temperature, or load.
5. **Accountability:** the engineer remains responsible for the design and its verification.

Project conclusion

The eight example circuits demonstrate how AI can support learning by explaining circuit operation, organising derivations, and suggesting simulator workflows. The project also shows that reliable results require independent hand calculations and actual simulator measurements. The best workflow combines AI for assistance with circuit theory, datasheets, and simulation for verification. AI should accelerate engineering work, not replace engineering judgement.

Future improvements

Extend the project with component tolerance sweeps, temperature variation, Monte Carlo analysis, real breadboard measurements, oscilloscope captures, and automated extraction of simulator measurements. Compare AI tools again when model versions change.

10. Presentation (6-slide outline)

Use this section as a ready-to-transfer slide script for a 5–7 slide presentation. Add your name, circuit screenshots, and actual tool scores.

Slide 1 — Title and problem

AI Circuit Explainer and Verification Studio
Project question: Can AI explain and analyse basic analog circuits, and how can answers be verified?

Slide 2 — Method

Choose eight circuits → ask four AI tools → calculate by hand → simulate in LTspice/Falstad → compare → log errors.

Slide 3 — Circuits studied

Voltage divider; RC low-pass; RC high-pass; BJT common-emitter amplifier; inverting op-amp; bridge rectifier; diode clipper; Zener regulator.

Slide 4 — Key numerical examples

Divider: 6.00 V. RC cutoff: 1.59 kHz. Inverting op-amp: gain −10 and output −2.00 V. Zener regulator: expected 5.10 V under stated load.

Slide 5 — Verification and AI comparison

Show the hand-calculation vs simulator table and the four-tool scoring chart. Include one genuine example of an AI error and its correction.

Slide 6 — Conclusion

AI helps explain and organise analysis, but assumptions, units, device models, and measurements must be checked. Engineering verification remains essential.

11. Final submission checklist and evidence pages

Complete these items before submitting the project. The report is designed to hold the written work; the actual simulator evidence must come from your own LTspice, Falstad, Tinkercad, or other approved simulator session.

Required final submission

■ Verified explanations for all eight circuits.
■ Hand calculations with units and assumptions.
■ Actual simulator screenshots for each circuit (schematic plus relevant plot or operating point).
■ Completed calculation-versus-simulation table with measured values and percentage errors.
■ Completed error log with observed AI mistakes and corrections.
■ Four-tool comparison with model/version, test date, prompts, and scores.
■ Prompt library and saved AI responses.
■ Study guide and limitations discussion.
■ Automation script and sample output CSV.
■ 5–7 slide presentation.
■ Final demonstration of one circuit from prompt through verification.

Simulator screenshot register

Circuit	Schematic screenshot attached	Plot / operating point attached	Measured result entered
Voltage divider	■ Yes / ■ No	■ Yes / ■ No	■ Yes / ■ No
RC low-pass	■ Yes / ■ No	■ Yes / ■ No	■ Yes / ■ No
RC high-pass	■ Yes / ■ No	■ Yes / ■ No	■ Yes / ■ No
BJT amplifier	■ Yes / ■ No	■ Yes / ■ No	■ Yes / ■ No
Inverting op-amp	■ Yes / ■ No	■ Yes / ■ No	■ Yes / ■ No
Bridge rectifier	■ Yes / ■ No	■ Yes / ■ No	■ Yes / ■ No
Diode clipper	■ Yes / ■ No	■ Yes / ■ No	■ Yes / ■ No
Zener regulator	■ Yes / ■ No	■ Yes / ■ No	■ Yes / ■ No

Final demonstration script (about 2 minutes)

1. Introduce the selected circuit and its purpose.
2. Show the exact component values and the AI prompt used.
3. Present the AI explanation and independent hand calculation.
4. Open the simulator schematic and waveform or operating-point results.
5. Compare values, calculate error, and explain any difference.
6. State one AI limitation and how engineering verification addressed it.

References and learning resources

Use the official documentation and datasheets for the exact devices and simulator version used. Suggested resources: LTspice official documentation; Falstad Circuit Simulator; manufacturer datasheets for the selected transistor, op-amp, rectifier diodes, and Zener diode; and a standard introductory electronic-circuits textbook. Add the exact URLs, datasheet part numbers, access dates, and textbook edition required by your institution.

Submission note: This PDF completes the written report structure and provides reproducible calculations and test instructions. It intentionally does not invent AI test scores or simulator screenshots. Fill in the evidence fields using your own tool runs before presenting it as a fully verified experimental report.